

Roll No.

TOE-713

B. TECH. (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

ARTIFICIAL INTELLIGENCE AND PATTERN RECOGNITION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define the term artificial intelligence. How is it different from natural intelligence ? (CO1)

OR

- (b) What is the difference between knowledge representation and knowledge acquisition ? (CO1, CO5)

2. (a) Discuss Issues in the design of Search Programs. (CO1, CO4)

OR

- (b) Explain the difference between Procedural and Declarative Knowledge. (CO4)

3. (a) Differentiate between Forward and Backward Reasoning in AI. (CO1)

P. T. O.

(2)

OR

(b) Elaborate the term matching in AI. Explain various matching techniques in AI. (CO2)

4. (a) Discuss Bayes Theorem in detail. (CO2, CO3)

OR

(b) Write the concept of Bayesian Inference with a suitable example. (CO2)

5. (a) Define Agents in AI. Give the structure and characteristics of Agents in AI. (CO1)

OR

(b) Elaborate the Heuristic Search Technique in Artificial Intelligence.

(CO2)

Roll No.

TOE-714

B. TECH. (CSE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

ARTIFICIAL INTELLIGENCE

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Discuss the steps involved in natural language processing with an example. (CO1)

OR

- (b) What is Strong AI, and how is it different from the Weak AI ? Give a brief introduction to the Turing test in AI. (CO1)

2. (a) Describe in detail the differences between language understanding and language generation. (CO3)

OR

- (b) Enumerate the Classical "Water Jug problem". Describe the state space for this problem. Solve this problem by giving its opera loss sequence.

(CO3)

P. T. O.

(2)

3. (a) Explain intelligence and artificial intelligence. How do you distinguish between the two ? Write a short note on Turing Test. (CO1/CO2)

OR

- (b) Explain AI Agents design philosophy using neat and clean diagrams. (CO1/CO2)

4. (a) Explain TOP-DOWN and BOTTOM-UP parsing in detail with the help of suitable example. (CO2/CO3)

OR

- (b) Discuss the following search technique with the help of an example. Also discuss the benefits and shortcoming of each.

(i) Depth First Search.

(ii) Breadth First Search. (CO2/CO3)

5. (a) Elaborate the difference between heuristic algorithm versus solution guaranteed algorithms. (CO2/CO3)

OR

- (b) Explain transition Net, RTN, and ATN in NLP. (CO2/CO3)

Roll No.

TOE-703

B. TECH. (CSE) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

INTELLECTUAL PROPERTY RIGHTS FOR ENGINEERS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define :

(i) Intellectual Property

(ii) Intellectual Property Rights

(CO1)

OR

(b) Give salient features of Nature of IP.

(CO1)

2. (a) Delineate the constitutional aspects of IP.

(CO3)

OR

(b) Showing branching hierarchy, show different creations covered under IP.

(CO2)

3. (a) Define the following terms giving *one* example each :

(CO1)

(i) Industrial Design

(ii) Patent

P. T. O.

(2)

OR

(b) Explain the following terms (any *two*) with one example : (CO2)

(i) Trademark

(ii) Trade secret

(iii) Geographical Indicator

4. (a) Classify IP rights showing these as Registered and Unregistered categories. State and explain various types of Patents. (CO2)

OR

(b) List some practical vitae point to consider in planning Start-up. (CO1)

5. (a) What are salient features of inventions that deem it patentable ? Give exceptions for these conditions. (CO3)

OR

(b) Illustrate real brands in order to showcase various aspects of Trademarks. What are various elements covered by Trade secrets ?

(CO2)

Roll No.

TCS-703

**B. TECH. (CSE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

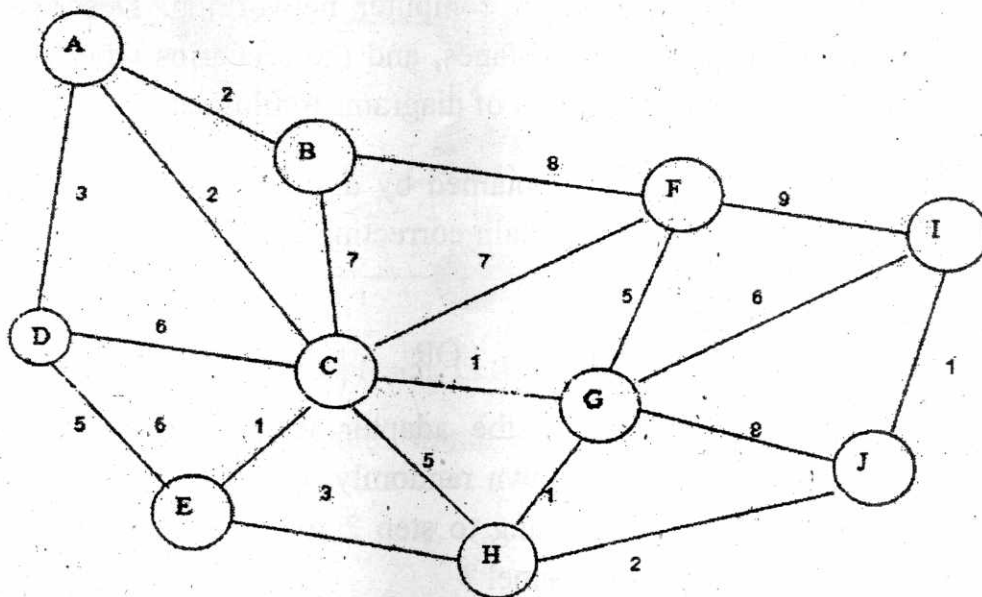
COMPUTER NETWORKS-II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

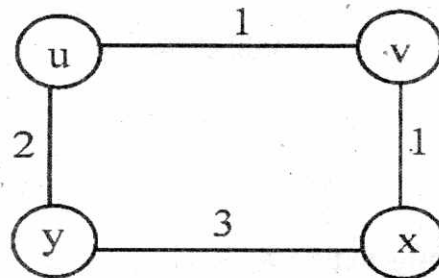
1. (a) Consider the following network. With the indication link costs, use LS Algorithm to compute the shortest path from 'D' to all network nodes. Show the result in the tabular format : (CO1)



P. T. O.

OR

- (b) Explain the key concepts and operation of the RIP (Routing Information Protocol) in computer networking. Provide insights into how RIP maintains routing tables, its limitation and potential scenarios where RIP might be suitable or unsuitable. Illustrate your answer with examples of diagrams as needed. (CO3)
2. (a) Consider the network shown below, and assume that each node initially knows the cost to each of its neighbours. Consider the distance-vector algorithm and show the distance table entries at node U. (CO1)



- (b) Explain the key concept and operation of the OSPF (Open shortest Path First) routing protocol in computer networking. Describe the OSPF routing process, its advantages, and the scenarios where OSPF is not suitable. Provide examples of diagrams to illustrate your answer. (CO2)
3. (a) What is the remainder obtained by dividing $x^7 + x^5 + 1$ by the generator polynomial $x^3 + 1$ and explain correcting single bit error using single bit parity? (CO1)

OR

- (b) In CSMA/CD protocol, the adapter waits $K \cdot 512$ bit times after a collision, where K is drawn randomly. For $K=100$, how long does the adapter wait until returning to step 2 for a 10 Mbps Ethernet? For a 100 Mbps broadcast channel? (CO2)

(3)

4. (a) Describe the role of all the field present in Ethernet frame and what is the actual reason for limiting the Ethernet payload to 1500 bytes ?

(CO2)

OR

- (b) Explain the role of a router in computer network. How does a router differ from a switch in term of functionality and the layer of the OSI model they operate in ?

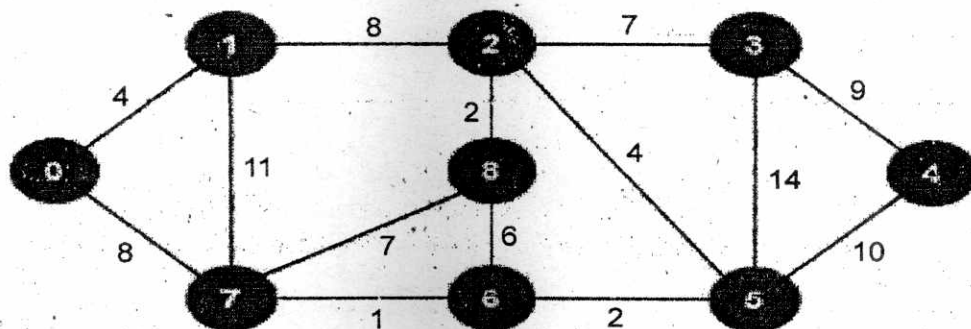
(CO3)

5. (a) A group of N stations share a 56 Kbps pure ALOHA channel. Each station outputs a 1000 bit frame on a average of once every 100 sec, even if the previous one has not yet been send (e.g. the stations are buffered) What is the maximum value of N ?

(CO2)

OR

- (b) (i) Design the spanning-tree for broadcasting for the network shown below :



- (ii) Explain all the fields present in the Ethernet Frame Structure.

- (iii) Calculate the maximum frame rate a node on Ethernet (assume 10 Mbps link)

(CO3)

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TCS-704

B. TECH. (CSE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

ADVANCED COMPUTER ARCHITECTURE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explore the key technological advancements and trends that have contributed to the significant improvements in processor performance.

(CO1)

OR

- (b) Examine the adaptation and diversification strategies employed by the semiconductor industry in response to the challenges to Moore's Law. Highlight the emergence of multi-core processors and innovations in memory hierarchies as ways to maintain performance gains.

(CO1)

P. T. O.

2. (a) Suppose a semiconductor manufacturer produces silicon wafers with a defect density of 0.08 defects per square centimetre. Each wafer has a surface area of 150 square centimetres. The facility produces 500 wafers per production run. If the cost of manufacturing a single wafer is \$2,000, and each wafer contains 1,000 die units, calculate :
- (i) The yield for a single wafer.
 - (ii) The total cost for the production run.
 - (iii) The cost per die. (CO1)

OR

- (b) Elaborate the core concept of Amdahl's Law, which relates to the potential speedup of a program when only a portion of it can be parallelized. Derive the mathematical expression of Amdahl's Law.

(CO1)

3. (a) Finding Execution Time for given CPI and Total Instructions. Consider a computer system with a CPI of 1.5 and a program that executes 10,000 instructions. Calculate the total execution time for this program.

(CO1)

OR

- (b) Examine the role of memory hierarchy in computer architecture. How has the design of memory subsystems evolved over time to bridge the speed gap between processors and memory, and what trade-offs have architects faced in this regard ?

(CO2)

4. (a) Compare and contrast three cache mapping techniques in terms of cache hit rates, hardware complexity, and their suitability for different computing workloads. Explain with the help of the diagram. (CO2)

OR

- (b) Consider a 16-way set-associative cache :

- (i) Data words are 64 bits long
- (ii) Words are addressed to the half-word
- (iii) The cache holds 2 Mbytes of data
- (iv) Each block holds 16 data words
- (v) Physical addresses are 64 bits long

How many bits of tag, index, and offset are needed to support references to this cache ? (CO2)

5. (a) Explore the key components of virtual memory systems, including the Page Table, Page Table Entry (PTE), and the Translation Lookaside Buffer (TLB). Explain how the Page Table maps virtual addresses to physical addresses and the role of TLB in caching frequently used translations. Discuss the advantages of TLB in improving memory access latency. (CO2)

OR

- (b) Consider a cache memory with the following specifications :

Cache size : 64 kB (kilobytes)

Block size : 32 bytes

Associativity : 4-way set associative

Memory address size : 32 bits

P. T. O.

You are given a decimal memory address : 245,678.

Calculate the following :

- (i) The number of sets in the cache.
- (ii) The number of lines (slots) per set.
- (iii) The size of the index field, offset field, and tag field in bits.
- (iv) The decimal values of the index, offset, and tag for the given memory address.

(CO2)

Roll No.

TOE-707

B.TECH. (CSE/SPECIALIZATION) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

CRYPTOGRAPHY AND NETWORK SECURITY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Classify security attacks involved in data communication over Internet.

(CO1)

OR

(b) (i) Eva has acquired the ciphertext that Alice has secretly sent to the Bob. Classify the attacks in the following scenarios and also justify your answer :

(I) Eva wants to retrieve the original plaintext by applying all possible key combinations.

(II) Eva is analysing the different cipher text to predict the plaintext.

(ii) Differentiate between a block cipher and a stream cipher. (CO1)

P. T. O.

2. (a) Use Playfair cipher to encrypt the message "Instrument", with the key "Encryption". (CO1)

OR

- (b) Encrypt the given plaintext by using the hill cipher. (CO1)

Plaintext : MOREMONEY

17 17 5
Key: 21 18 21
2 2 19

3. (a) Discuss various transformation performed in AES. (CO2)

OR

- (b) Explain various modes of operations for a block cipher. (CO2)

4. (a) (i) In the context of statistical cryptanalysis, what do you understand by the terms diffusion and confusion, and how they are introduced in block cipher?

- (ii) Differentiate between random and pseudo-random numbers.

(CO2/CO3)

OR

- (b) (i) Discuss the criteria defined by NIST for selecting the AES.

- (ii) Explain the linear congruential generator for generating pseudo random numbers. Further exemplify the importance of selecting the values a, c and m in the same. (CO2/CO3)

(3)

5. (a) Analyse and explain the process of key distribution for two parties A and B via KDC, where A and B both share a secret key with KDC. (CO3)

OR

- (b) Discuss Blum Blum Shub generator approach to generate pseudo-random number. Generate the random number sequence x_1, x_2, x_3, x_4 by using the Blum Blum Shub generator where $p = 11, q = 19$, and $x_0 = 100$. (CO3)

Roll No.

TIT-721

**B. TECH. (CSE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

BUSINESS INTELLIGENCE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are some of the automated IT Applications that are a great help to your university in its day-to-day functioning ? Explain. (CO1)

OR

- (b) Explain about Malcolm Baldrige Criteria framework for any organization. (CO1)

2. (a) What are processes involved in the Business Process Management ?

(CO1)

OR

- (b) What are the key components of Malcolm Baldrige Criteria for performance excellence framework ? (CO1)

P. T. O.

(2)

3. (a) What is your understanding of the connected world ? Discuss about the Internet-ready applications advanced common capabilities. (CO1)

OR

- (b) Discuss best practices for managing the growth of unstructured data. (CO2)

4. (a) Web pages are said to be unstructured data even though they are defined by the HTML markup language which has a rich structure. Why ? (CO2)

OR

- (b) Can XML data be converted into a structured format ? Explain with an example. (CO2)

5. (a) What according to you is the impact of unstructured data on backup and recovery ? (CO2)

OR

- (b) How is OLTP different from ERP ? Write some queries that an OLTP system can process easily. (CO3)

Roll No.

TCS 731

B. TECH. (CSE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
COMPUTER FORENSICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) What are the various tools and softwares used in a cyber forensic investigation ? (CO2)

OR

- (b) Illustrate the digital forensic investigation model. Discuss each of the steps involved while investigating a digital forensic case. (CO3)
2. (a) Compare and contrast cybercrimes in India with cybercrimes in the rest of the world, highlighting key differences and similarities. (CO1)

OR

- (b) Discuss the methods and best practices for collecting digital evidence during an investigation. (CO2)

P. T. O.

(2)

3. (a) How can deleted evidence be recovered in a cyber forensics investigation ? Provide examples of tools and techniques. (CO4)

OR

- (b) Write short notes on any *two* of the following : (CO4)

(i) Multimedia forensics

(ii) Virtual Machine forensics

(iii) Database forensics

4. (a) Analyze the advantages and disadvantages of adopting digital forensics as a field of investigation. (CO1)

OR

- (b) Provide an overview of memory forensics and its role in analyzing digital devices for evidence. (CO3)

5. (a) What is the "Chain of Custody," and why is it important in computer forensics ? (CO3)

OR

- (b) Trace the history and evolution of digital forensics as a field of study. (CO5)

Roll No.

TCS-771/TCS-742

B. TECH. (CSE) (SEVENTH SEMESTER) MID SEMESTER EXAMINATION, Oct., 2023

NATURAL LANGUAGE PROCESSING WITH BIG DATA

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Explain ambiguity in natural language. Why is it a challenge in natural language processing ? Explain with the help of any *two* examples.

(CO1)

OR

- (b) Using an example, illustrate the difference between stemming and lemmatization in natural language processing. (CO1)
2. (a) In TF-IDF weights calculation why is logarithm used ? Given below is a corpus of three documents, compute the TF-IDF weight of the term display in Document 2. You need to just fill in the numbers in the formula for TF-IDF (no need to finish calculation). (CO2)

Document 1 : *this phone model has good amoled display*

Document 2 : *the display of tv is broken but the phone has good display*

Document 3 : *Indian foreign minister is travelling*

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(2)

OR

- (b) How does smoothing help in dealing with rare words in natural language processing ? Explain with the help of any *one* example. (CO1)
3. (a) Compute cosine similarity between the following two sentences : (CO2)
- Sentence 1 : *natural language processing subject in seventh semester*
- Sentence 2 : *seventh semester subject in graphic era*

OR

- (b) How Naïve Bayes classifier distinguishes spam text from ham ? Illustrate with the help of an example. (CO2)
4. (a) Explain bag of words model. Show how bag of words model will be used in the following phrase (assume no stop word removal) : she sells sea shells on the sea shore. (CO1)

OR

- (b) Describe the working of skip gram with negative sampling (SGNS) in computing word2vec embeddings. (CO2)
5. (a) Compute Precision, Recall, F1, and accuracy of a text classifier, given the following confusion matrix : (CO2)

	Predicted Positive	Predicted negative
Actual Positive	45	15
Actual negative	24	32

OR

- (b) Assume you have a dataset of 2000 samples. Explain how to perform 10-fold cross-validation with stratified sampling on this dataset. (CO2)

Roll No.

TCS-750

B. TECH. (CSE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
CLOUD ORCHESTRATION AND LOAD BALANCING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Healthcare systems around the world are unsustainable. The global pandemic has only exacerbated existing staffing and physician shortages, increased cost pressures, and increased service backlog and capacity issues. For global virtual care company, Teladoc Health, the partnership between HITRUST and Microsoft enables data security, ensures regulatory compliance, and helps their platform maintain and grow trust with partners and patients. Evaluate Cloud automation from perspective of above mentioned organization. (CO1)

OR

- (b) When Provincial Health Services Authority (PHSA) began seeing the need to deliver faster, more comprehensive health insights to healthcare workers across British Columbia, it took action. Old, unstandardized requests were slowing data analysis and the sharing of insights. To

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standardize and speed data access, PHSA moved its healthcare analytics platform to Microsoft Azure, creating a readily accessible data analysis platform that is scalable and secure. Data requests that once took many months now take only days. Because of the sensitive nature of health data, PHSA has worked closely with Microsoft to make sure health data never leaves Canada and is kept highly secure. Evaluate Google Cloud Load Balancing from perspective of above mentioned organization.

(CO2)

2. (a) Hamad Medical Corporation (HMC) is the leading healthcare provider in Qatar. It was looking for a security information and event management (SIEM) solution. "Our vision was to deliver security without compromising access and usability. We also wanted to ensure patient and corporate confidentiality at all times," says Richard Storey, Executive Director/Acting Chief of Health Information and Communication Technology at HMC. With Microsoft Sentinel, HMC gained 360°, real-time visibility over cybersecurity, making proactive responses faster for supporting staff to access information. Analyze AWS Lambda function from perspective of above mentioned organization.

(CO2)

OR

- (b) Growing healthcare organization MultiCare Health System teamed up with Microsoft and Microsoft partner 3Cloud to meet changing demands in health data access prompted by the use of fitness trackers and phones. MultiCare wanted a scalable, interoperable data exchange framework to connect providers, patients, and payers—key for success in population health and value-based care. It also required an adaptable

solution. Working with Microsoft FHIR experts and partnering with 3Cloud, MultiCare built a unique, fully managed solution using Microsoft Azure. The results have surpassed expectations : in three months, MultiCare has seen a significant reduction in administrative burden with a \$50 per-chart cost savings and a 65 percent reduction in labor. Analyze the AWS CloudFormation from the perspective of above mentioned organization. (CO3)

3. (a) Mercy, one of the largest healthcare systems in the United States, aided its digital transformation by using Microsoft Azure to decrease operational costs, improve the patient experience, and give its providers better access to data. The company wanted to modernize its infrastructure and extract more value from its sizeable data archives to keep up with evolving patient and doctor expectations. With support from Microsoft, Mercy identified business outcomes and undertook a digital transformation to achieve them. On Azure, the company improved efficiency, connected previously siloed data, reduced patient stay durations, and built a foundation for ongoing innovation. Analyze IBM Cloud Orchestrator from the perspective of above mentioned organization. (CO2)

OR

- (b) When Centro Hospitalar de Trás-os-Montes e Alto Douro needed a way to protect sensitive, highly confidential patient data and applications in its hybrid environment, it chose Microsoft Security solutions. The hospital system's future-forward strategy calls for a dynamic, cloud-based security approach that will stay on top of threats and position the

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organization for cloud migration. It now has a centralized view of security events with Microsoft Sentinel, increasing frontline defense with Azure Automation and Microsoft Defender for Cloud. With Azure Arc to extend Centro Hospitalar's Azure platform protections to its on-premises assets, the hospital system considers its IT security vaccinations up to date. Analyze RedHat Ansible from the perspective of above mentioned organization. (CO3)

4. (a) CooperSurgical, a leading fertility company, adopted Microsoft Dynamics 365 as a unified CRM solution to deliver transparency during a time of rapid global growth. The company nearly doubled in size in two years, making it difficult to get an accurate overview of operations and opportunities. Many recent acquisitions were using their own CRM and ERP systems, and knowledge sharing was a time-consuming process. CooperSurgical decided to adopt Microsoft solutions globally to enhance forecasting, relationship management, and productivity. Now, the company is poised to bring healthcare products and services to patients more quickly than ever. Analyze Azure Automation from the perspective of above mentioned organization. (CO3)

OR

- (b) The Mount Sinai Health System (Mount Sinai), an integrated medical system, is innovating at the leading edge of medical care using AI solutions on Microsoft Azure. Since launching its Clinical Data Science (CDS) initiative in 2015, Mount Sinai has improved care through analytics and the creation of tools that predict the needs of patients. In 2021, it migrated the CDS infrastructure to Azure to rescale compute

capacity and better meet current and future needs. Through its strategic partnership with Microsoft, Mount Sinai is accelerating its use of AI innovation to identify risks to patients, including malnutrition, delirium, and falls. Now, hospital stays have been reduced, mortality rates are falling, and providers are engaged. Analyze Multi-Cloud Orchestrator from the perspective of above mentioned organization. (CO2)

5. (a) San Raffaele University and Research Hospital is one of the largest of its kind in Italy, as well as a pioneer of cutting-edge translational research and technology adoption at both a national and international level. During the pandemic, the organization saw an opportunity to start using artificial intelligence to better triage incoming symptomatic patients and identify their future reaction to the virus. Fast forward three years, the pilot project has been turned into an intelligent data solution powered by AI - which San Raffaele is going to use to transform clinical research, improve the end-to-end patient journey inside the hospital and accelerate its own move towards precision medicine. Analyze the AWS Systems Manager Automation from the perspective of above mentioned organization. (CO2)

OR

- (b) Health First, a fully integrated delivery network in Florida, created a data management solution by joining its existing data warehouse architecture with Azure Synapse Analytics. While collecting information from more than 60 applications, Health First faced challenges in scaling its on-premises data warehouse. Additionally, the

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health network wanted to explore AI, predictive analytics, and other innovative technologies while maintaining a high level of its data security for patient data. It deployed Azure Synapse Analytics to scale integration and storage capabilities in a secure cloud system. By creating this system, Health First lowered costs, improved workload processing times, and offered better data access for providers. Analyze OpenStack Heat orchestration engine from the perspective of above mentioned organization.

(CO3)

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Roll No.

TCS-756/TOE-701

B. TECH. (ALL-CSE BRANCH) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

HUMAN COMPUTER INTERACTION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What qualities define a successful interface ? Identify the human factors that must be taken into account for the design of an HCI with an appropriate diagram. (CO3)

OR

- (b) What factors are taken into account when determining the speed of human interaction ? One by one, explain. (CO2)

2. (a) What human factors must to be taken into account while designing screens ? Explain each factor listed in your response. (CO2)

OR

- (b) Comparison of an AI-enabled user interface with a normal user interface (without AI), using an appropriate example. (CO2)

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(2)

3. (a) A word processor or web browser's vertical scroll bar is one example. Give one example of how the scrollbar uses each of the five principles listed below by Norman for good design. In a single statement, explain the scrollbar's interface model. (CO3)

OR

- (b) Explain the development of the graphical user interface in chronological order. (CO2)
4. (a) Explain HCI. Talk about the advancements and significance of the audio-visual interface as a tool for enhancing human user productivity. (CO1)

OR

- (b) Explain the steps involved in designing a web interface. (CO1)
5. (a) What are the benefits and drawbacks of a graphical system? (CO1)
- OR
- (b) Define good screen design. What are the problems with poor design given the factors? (CO1)

Roll No.

TCS-761

B. TECH. (CSE)+ALL/B. TECH. (BIOTECH.)

(SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

CLOUD INFRASTRUCTURE SERVICES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) By 2025, organizations will be deploying 95% of all new workloads on cloud-native platforms. Specify the driving force behind migration towards cloud computing in large scale. (CO1)

OR

- (b) Explain the difference between client server model and cloud computing. (CO1)

2. (a) Explain the various cloud service models with respect to cloud user control. (CO3)

OR

- (b) Explain the Layered Architecture of Cloud. (CO3)

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(2)

3. (a) Explain the steps involved in cloud computing life cycle management implementation. (CO3)

OR

- (b) List the management responsibilities for administration of clouds. (CO3)

4. (a) Explain the Cloud Infrastructure Management Interface model. (CO2)

OR

- (b) Explain the DMTF standard for CIMI. (CO2)

5. (a) Discuss the challenges need to addressed for implementing Infrastructure Security at application Level in Cloud Computing. (CO3)

OR

- (b) Compare and contrast DoS and EDoS attacks in cloud computing. (CO3)

Roll No.

TCS-785

**B. TECH. (CSE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

DEEP LEARNING FOR COMPUTER VISION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Why digital image processing steps are required in computer vision ?
Explain with the help of a real-time image processing application.

(CO1)

OR

- (b) Consider two color images have the same histogram. Write your comments on the following and justify your answer with suitable example :

(CO1)

- (i) Is the resolution of both images equal or not ?
- (ii) Are the contrast and brightness of both images same or different ?
- (iii) Is the total storage size equal or not ?
- (iv) Are both input images same ?

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2. (a) How 6-bit gray-scale image can be converted into an equivalent of the 4-bit gray-scale image ? Explain and justify your answer with a suitable example. (CO1)

OR

- (b) How much storage capacity is required to store image with size of 512×384 and 128 gray levels ? Also identify the number of bits required to store the pixel value. (CO1)

3. (a) Explain how the Log Transformation and Power-Law Transformation are used to enhance the images. Explain each and every case with the help of MATLAB code. (CO2)

OR

- (b) Suppose that a 3-bit image ($L = 8$) of size 5×5 pixels as shown in below matrix, where the intensity levels are integers in the range $[0, L - 1]$:

$$\begin{bmatrix} 6 & 5 & 4 & 3 & 2 & 12 \\ 5 & 4 & 3 & 2 & 12 & 2 \\ 4 & 3 & 2 & 12 & 2 & 3 \\ 3 & 2 & 12 & 2 & 3 & 4 \\ 2 & 5 & 4 & 3 & 2 & 3 \\ 12 & 4 & 3 & 2 & 12 & 4 \end{bmatrix}$$

- (i) Compute the histogram for given matrix.
(ii) Calculate transformation function of histogram.
(iii) Compute the equalized histogram. (CO2)
4. (a) How smoothing filter works in image processing ? Which method is effectively works for image smoothness in spatial domain ? Apply that method by considering a 5×5 image matrix for enhancing smoothness in the images. (CO2)

(3)

OR

- (b) Why are sharpening filters required in image processing ? Which method is effectively works for image sharpness in spatial domain ? Apply that method by considering a 5×5 image matrix for enhancing sharpness in the images. (CO2)
5. (a) Why does the noise considered to be an additive in images ? How median filter works on the noisy image ? Explain each and every step by apply 3×3 median filter on below Image Matrix : (CO3)

$$\begin{bmatrix} 6 & 5 & 4 & 3 & 2 & 12 \\ 5 & 4 & 3 & 2 & 12 & 2 \\ 4 & 3 & 2 & 12 & 2 & 3 \\ 3 & 2 & 12 & 2 & 3 & 4 \\ 2 & 3 & 2 & 12 & 2 & 3 \\ 12 & 2 & 12 & 2 & 3 & 4 \end{bmatrix}$$

OR

- (b) Perform image segmentation using Otsu's thresholding for below matrix by considering a single threshold value : (CO3)

$$\begin{bmatrix} 6 & 5 & 4 & 3 & 2 & 12 \\ 5 & 4 & 3 & 2 & 12 & 2 \\ 4 & 3 & 2 & 12 & 2 & 3 \\ 3 & 2 & 12 & 5 & 4 & 3 \\ 2 & 5 & 4 & 4 & 3 & 2 \\ 12 & 4 & 3 & 3 & 2 & 12 \end{bmatrix}$$